

Datasets	Base Models	$\theta_2 \rightarrow \theta_1$ vs. $\theta_1 \rightarrow \theta_2$	$\theta_3 \rightarrow \theta_2$ vs. $\theta_2 \rightarrow \theta_3$	$\theta_3 \rightarrow \theta_1$ vs. $\theta_1 \rightarrow \theta_3$
Alpaca	Llama2-7B	0.1589 ↓ 0.2018 ↑	0.1953 ↓ 0.2143 ↑	0.1666 ↓ 0.2346 ↑
	Llama2-13B	0.1772 ↓ 0.1958 ↑	0.2149 ↓ 0.2408 ↑	0.1835 ↓ 0.2345 ↑
	Llama3-8B	0.2540 ↓ 0.2573 ↑	0.2268 ↓ 0.3229 ↑	0.2341 ↓ 0.2589 ↑
	Llama2-7B	0.1909 ↓ 0.2069 ↑	0.1719 ↓ 0.1721 ↑	0.2011 ↓ 0.2542 ↑
Truthful	Llama2-13B	0.1704 ↓ 0.1830 ↑	0.1544 ↓ 0.1640 ↑	0.1825 ↓ 0.2429 ↑
	Llama3-8B	0.2118 ↓ 0.2256 ↑	0.2100 ↓ 0.2173 ↑	0.2393 ↓ 0.2898 ↑
	Llama2-7B	0.2730 ↓ 0.2809 ↑	0.2654 ↓ 0.2691 ↑	0.2845 ↓ 0.2883 ↑
Safe	Llama2-13B	0.2419 ↓ 0.2439 ↑	0.2320 ↓ 0.2327 ↑	0.2464 ↓ 0.2606 ↑
	Llama3-8B	0.2097 ↓ 0.2156 ↑	0.2008 ↓ 0.2427 ↑	0.2277 ↓ 0.2709 ↑

Table 1: **Comparison between *inverse alignment* and *forward alignment* training loss.** Under different model, task, and stage slicing settings, *forward alignment* is more challenging than *inverse alignment*, i.e., the loss is higher. This indicates that pre-trained models tend to maintain their original distribution.

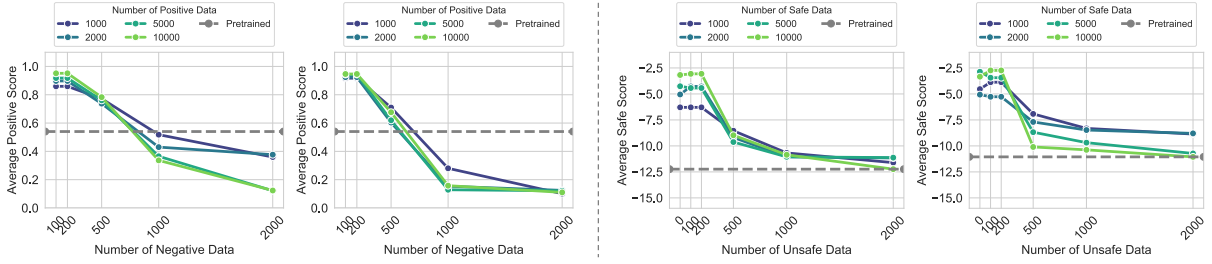


Figure 4: **Experimental results for validating the existence of rebound (left: IMDb, right: Beavertails).** The left part of each sub-figure is the performance of Gemma-2B while the right is Llama2-7B, respectively. Models trained with more **positive** data initially perform better but perform worse after fine-tuning with **negative** data.

the more **positive** the post-trained models’ performance, the more **negative** it becomes after *inverse alignment*. We validate tasks involving two opposing characteristics (e.g. safe and unsafe). We first train slices $\{\theta_1, \theta_2, \dots, \theta_n\}$ based on a pre-trained model θ_0 using positive (e.g., safe) data of various volumes. Then we perform inverse finetuning on these slices using negative data (i.e., unsafe).

Tasks and Datasets. We select two tasks: positive generation and single-turn safe conversation. For the former, we use the data classified as positive or negative in the IMDb dataset (Maas et al., 2011). Referring to (Rafailov et al., 2024), we use the first 2-8 tokens of each complete text as a prompt for LLMs to generate the subsequent content. For the latter, we use the data classified as safe and unsafe in Beavertails (Ji et al., 2024c). We organized the positive sample sizes into $\{1000, 2000, 5000, 10000\}$, while the negative sample sizes, were divided into $\{100, 200, 500, 1000, 2000\}$.

Evaluation and Metrics. We collect the model’s responses on the reserved test prompts. Then we use score models provided by existing research to complete the performance evaluation. For positive style generation, we refer to (Rafailov et al., 2024)

and use the Sentiment Roberta model (Hartmann et al., 2023) to classify the responses, taking the proportion of all responses classified as positive as the model score. For single-turn safe dialogue, we use the cost model (Dai et al., 2024) to score the safety of each response, using the average score of all responses as the model score.

Experiment Results. We evaluate the *rebound* phenomenon on Llama2-7B (Touvron et al., 2023) and Gemma-2B (Team et al., 2024). The experimental results in Figure 4 show that, for models fine-tuned with a larger amount of positive sample data, their performance drops quicker under only a small amount of negative sample fine-tuning. Subsequently, the performance decline slows down and tends to stabilize. This result also confirms the previous conclusion: the initial rapid decline of model’s performance is due to *rebound*, as the model is far from the pre-trained distribution; while the later stabilization of the countermeasure is due to *resistance*, as the model is already close to the pre-trained distribution.

To assess the generalizability of the *rebound* phenomenon, we perform additional ablation studies focusing on alignment algorithms, evaluation

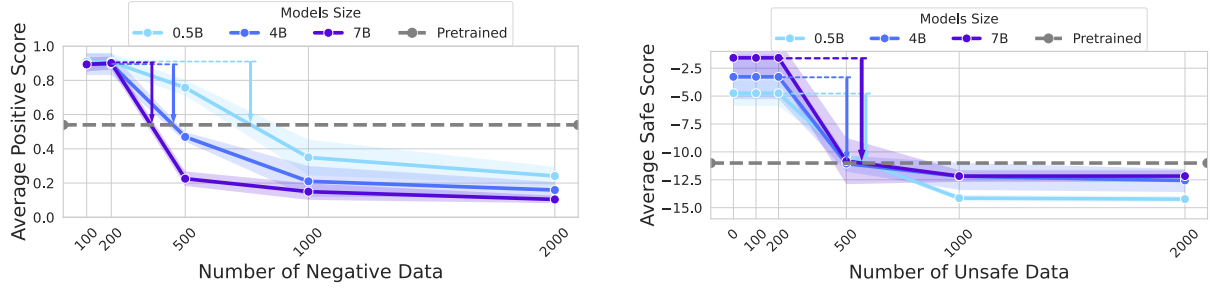


Figure 5: **Experimental results for validating *rebound* increases with model size (left: IMDb, right: Beavertails).** All single line covers positive data volume settings as Figure 4, with shadow denoting std. As the model size increases, the performance of the aligned model deteriorates more rapidly after fine-tuning with **negative** data.

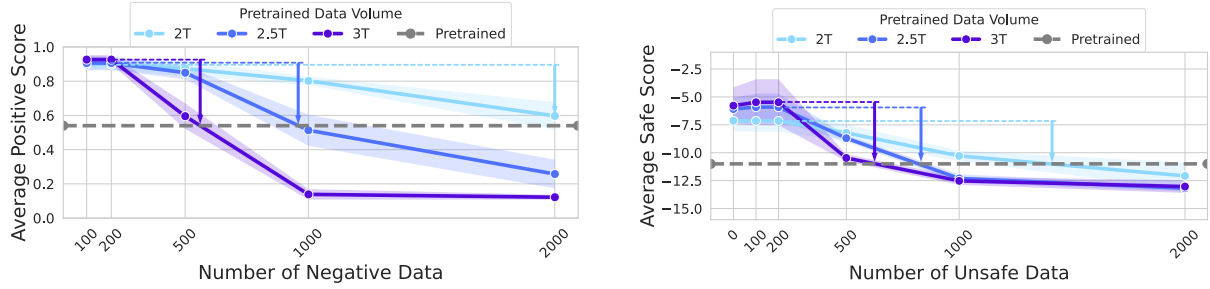


Figure 6: **Experimental results for validating *rebound* increases with pre-training data volume. (left: IMDb, right: Beavertails).** Each line covers positive data settings as Figure 4, with shadow denoting std. As pre-training data volume increases, aligned model performance deteriorates more rapidly after fine-tuning with **negative** data.

metrics, and fine-tuning directions. The results consistently validate the presence of the *rebound* phenomenon across language models. Further experimental details are available in Appendix B.2.

5.3 Internal Factor of *Rebound*

We conduct an in-depth analysis of the internal factors affecting the *rebound* phenomenon since it is crucial for the robustness of LLMs alignment. In particular, we investigate how model parameter scale and the amount of pre-training data influence *rebound*. Owing to space constraints, further experimental analyses are included in Appendix B.3.

Rebound Increases with Model Size. To investigate how the *rebound* phenomenon varies with model size, we conducted experiments on Qwen models (Bai et al., 2023) with parameter scales of 0.5B, 4B, and 7B. The experimental results in Figure 5 show that as the model parameter size increases, the initial performance decline due to negative data fine-tuning is faster, while the subsequent decline is slower. This indicates that as the parameter size increases, there is an increase in *rebound* in response to both positive and negative data, further suggesting a positive correlation between model *elasticity* and parameter scale.

Rebound Increases with Pre-training Data Volume. To verify that *rebound* increases with the growth of pre-training data, we vary pre-training slices (2.0T, 2.5T, and 3.0T) released by TinyLlama (Zhang et al., 2024a) and conduct the same experiment. As shown in Figure 6, when the pre-training data volume increases, the initial performance decline due to negative data fine-tuning is faster, while the subsequent decline is slower. It demonstrates that larger pre-training data volumes reinforce the *rebound* of LLMs, which is consistent with the inference proposed in Theorem 4.2.

6 Conclusion and Outlook

Our work uncovers a possible mechanism underlying the fragility of alignment: the *elasticity* of language models. We demonstrate that upon perturbation, the normalized compression rate of language models changes inversely with dataset size, making them more inclined to retain pre-training distributions while forgetting alignment distributions, thus resisting alignment. Through extensive experiments, we validate the universality of this *elasticity* effect, observing that the *elasticity* strengthens as model size and pre-training data volume scale up. Fundamentally, our findings support

a core assertion: language models exhibit *elasticity*, and thereby inherently resist alignment.

6.1 Limitations

Theory-wise, the primary limitation of our work is our specification of the *mass distribution* (Assumption A.7), and empirical studies on the exact form of this distribution shall be valuable. Experiment-wise, we have not systematically validated *elasticity* throughout the entire lifecycle of pre-training and alignment phases due to cost constraints. In future works, we plan to focus on whether this phenomenon is universally applicable, such as in multimodal models (Ji et al., 2024d; Huh et al., 2024). Additionally, we aim to theoretically uncover the relationship between model *elasticity* and *scaling laws* (Kaplan et al., 2020; Xiao et al., 2024), specifically determining the amount of training data required for *elasticity* to manifest and quantitatively analyzing whether *elasticity* intensifies as model parameters and pre-training data volume increase.

6.2 Broader Impacts

Rethinking Fine-tuning and Model Weight from Resist Alignment. Alignment fine-tuning methods aim to efficiently adjust the distribution of LLMs with minimal data to enhance their safety mechanisms. However, current alignment algorithms for language models may rely on optimization shortcuts, leading to local optima, while lacking involvement with the intrinsic mechanisms of the model (Qi et al., 2024a). From the perspective of language models’ *elasticity*, we require more robust alignment methods to ensure that modifications to model parameters extend beyond superficial changes (Qi et al., 2024a; Cohen et al., 2024), thereby preventing the emergence of more effective *inverse alignment* techniques that could give rise to extreme misalignment risks such as deceptive adversarial alignment (IDAIS-Beijing, 2024; Bengio et al., 2025; Marks et al., 2025; Baker et al., 2025) or alignment faking (Greenblatt et al., 2024). Although methods such as data cleansing during the training phase have proven to be an effective strategy for improving the malleability of a language model’s final distribution (He et al., 2024; Qi et al., 2024b), such approaches are not particularly cost-effective and feasible. In Appendix C.2, we provide a preliminary discussion of practical steps based on *elasticity* to mitigate *inverse alignment* risks, and we look forward to future research and practical developments grounded in *elasticity*, which

will enable the design of more robust alignment algorithms (Revel et al., 2024; Zhang et al., 2024b; Chen et al., 2024; Sheshadri et al., 2024; Liu et al., 2024; Li et al., 2025), ultimately achieving true and reliable alignment for LLMs.

Rethinking Open-sourcing from Elasticity of LLMs. Open-sourcing is a double-edged sword (Seger et al., 2023; Anwar et al., 2024; Kukreja et al., 2024). The public release of model weights enables the technical community to quickly identify and address potential vulnerabilities and conduct large-scale research on broad safety concerns (Touvron et al., 2023). These efforts collectively help sustain a balanced offence-defense dynamic (Jervis, 1978; Garfinkel and Dafoe, 2021) in the open-source environment and improve model security (Eiras et al., 2024). However, the risk of misuse associated with open-sourcing models is inevitable (Seger et al., 2023). Malicious fine-tuning of open-source models (Urbina et al., 2022; Sandbrink, 2023) and the use of the open-source models to facilitate system jailbreaks (Zou et al., 2023) can pose serious threats to public safety (Goldstein et al., 2023; Reuel et al., 2024). At present, the open-sourcing of powerful language models relies on safety alignment and rigorous security audits (Mökander et al., 2024) to ensure responsible use. However, if advanced *inverse alignment* techniques become feasible, even carefully aligned and audited open-source models could be reverted to their pre-alignment states at minimal cost, drastically lowering the barrier for jailbreaking aligned models and disrupting the offense-defense balance in the open-source ecosystem (Shapiro and Siegel, 2010). Therefore, it is crucial to develop alignment algorithms with fine-tuning robustness. We hope that deeper insights into the *elasticity* mechanisms of language models will drive the advancement of untunable alignment methods, enabling models to maintain reliable safety throughout their entire lifecycle (Madiaga, 2021).

References

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